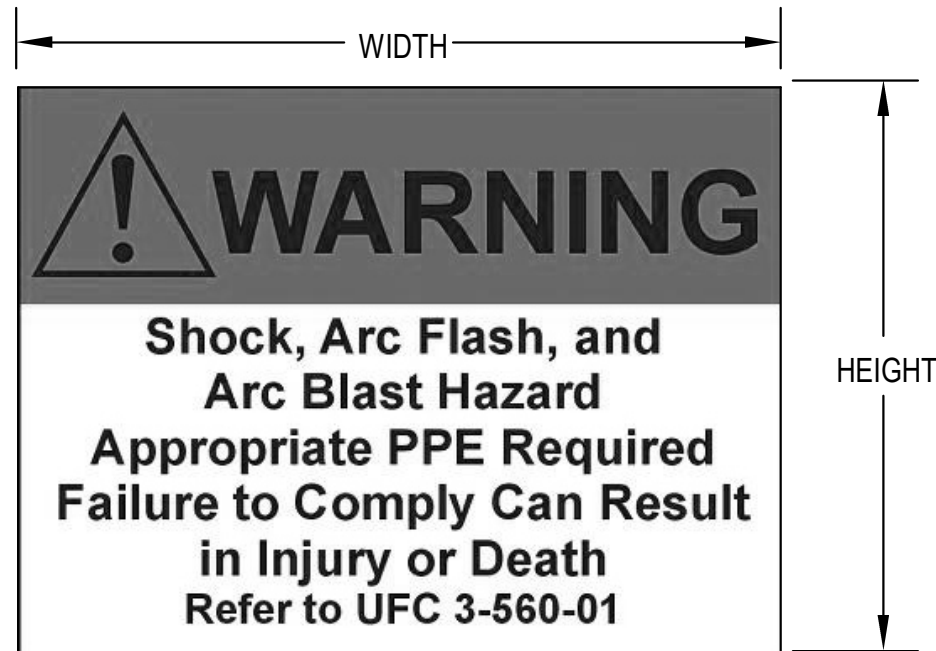


D

C

B

A



NOTES:

1. PROVIDE SELF-ADHESIVE VINYL LABEL TO AFFIX TO ELECTRICAL EQUIPMENT TO WARN OF ARC FLASH HAZARDS.
2. THE LABEL FORMAT AND TEXT SHALL BE IN ACCORDANCE WITH THE FIGURE.
3. THE LABEL SHALL BE LOCATED ON THE EQUIPMENT TO BE CLEARLY VISIBLE TO QUALIFIED PERSONS BEFORE EXAMINATION, ADJUSTMENT, SERVICING, OR MAINTENANCE OF THE EQUIPMENT.
4. THE SIZE OF THE LABEL SHALL BE MINIMUM:

EQUIPMENT TYPE	HEIGHT	WIDTH
INDOOR	2"	3"
OUTDOOR	3"	4.5"
5. A DOWNLOADABLE WINDOWS METAFILE IS AVAILABLE ON THE WHOLE BUILDING DESIGN GUIDE WEBSITE (WWW.WDBG.ORG) FOR USE IN A LABEL MAKING MACHINE.
 - A. THE FILE IS LOCATED ON THE "NAVFAC CADD DETAILS" PAGE. TO NAVIGATE TO THIS LOCATION, FOLLOW: HOME > DOCUMENTS & REFERENCES > CCB > CADD LIBRARY > NAVFAC CADD RESOURCES > NAVFAC CADD DETAILS.
 - B. ALTERNATIVELY, TYPE IN THE FOLLOWING ADDRESS IN INTERNET EXPLORER: HTTP://WWW.WDBG.ORG/CCB/BROWSE_CAT.PHP?C=232

GENERAL ARC FLASH WARNING LABEL

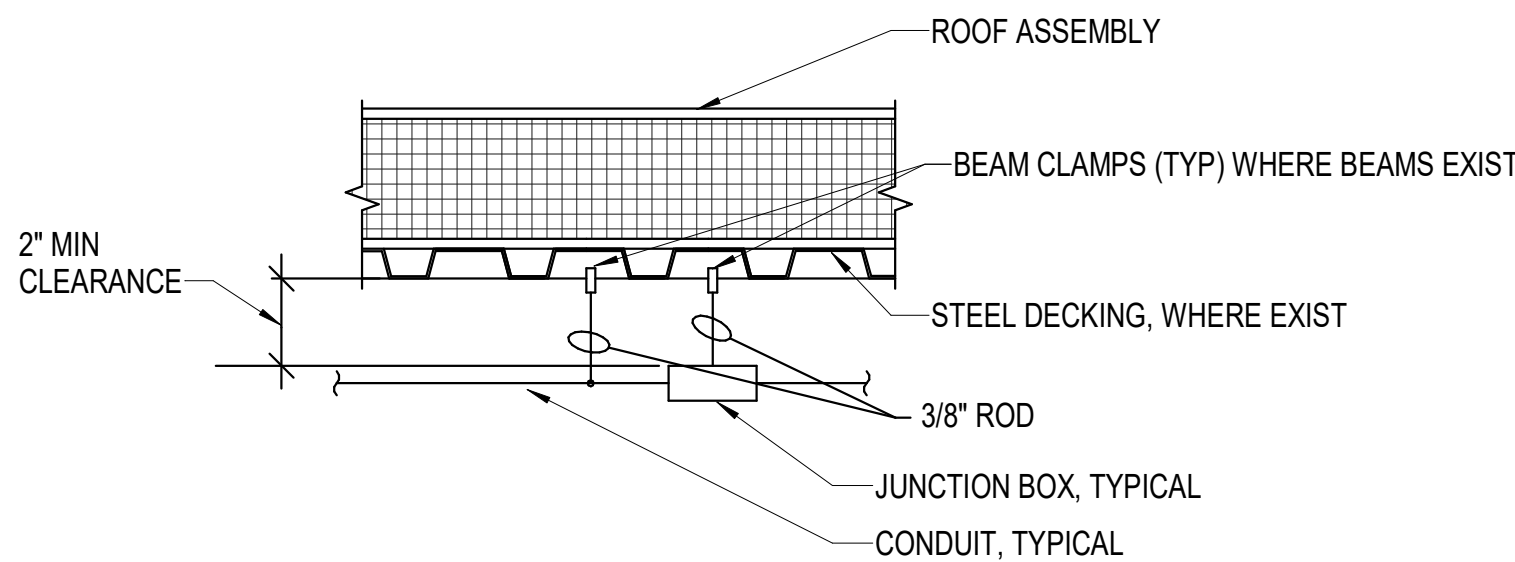
SKETCH DATE	APRIL 2015	STYLE	AF-1
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ARC FLASH WARNING LABEL DETAIL NOTE:

1. PROVIDE AT EACH PANELBOARD, DISCONNECT, STARTER, HVAC CONTROL PANEL, AND LIGHTING CONTROL PANEL.

B1 ARC FLASH LABEL

SCALE: NOT TO SCALE

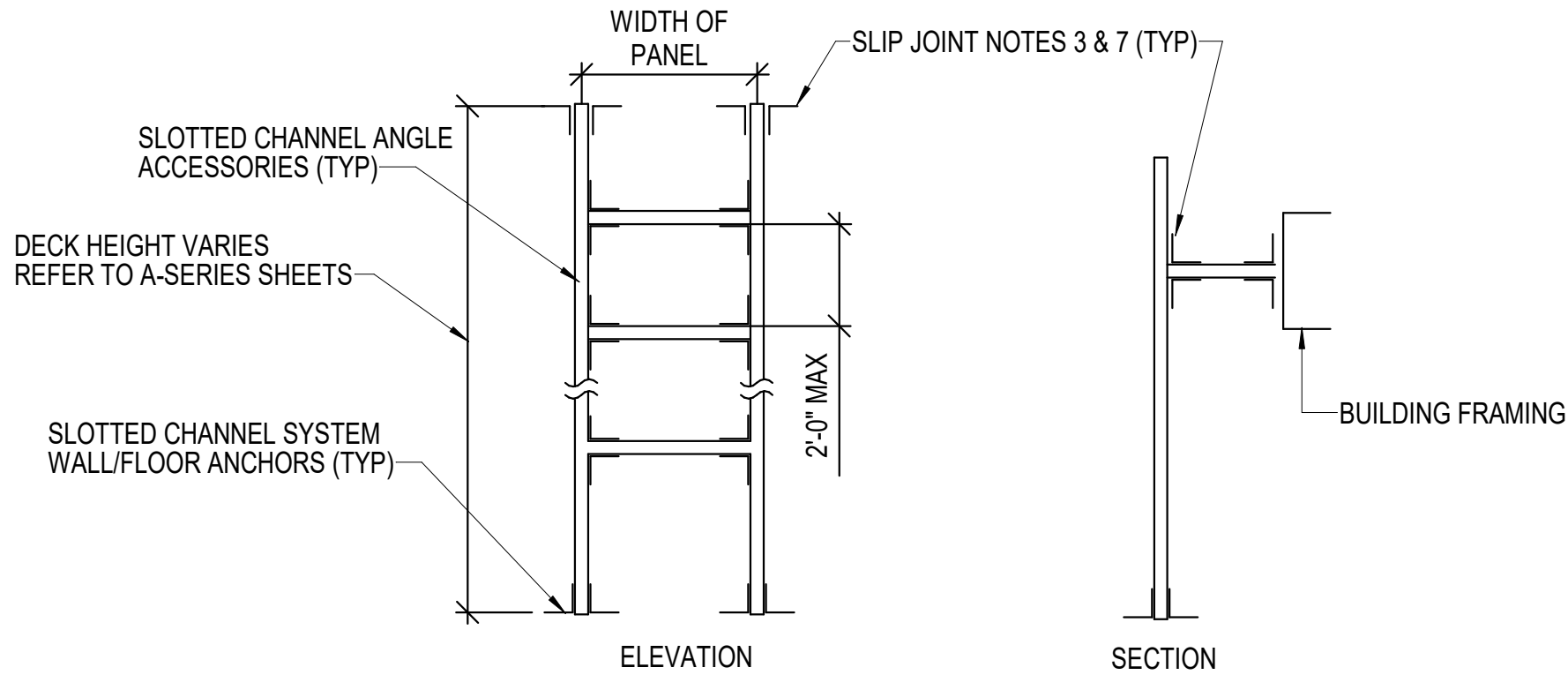


CONDUIT AND BOXES BELOW ROOF DECK INSTALLATION DETAIL NOTES

1. PROVIDE SUPPORT TO MAINTAIN MIN OF 2 INCHES OF CLEARANCE BETWEEN THE BOTTOM OF ROOF DECK AND THE TOP OF CONDUITS, BOXES, FIXTURES, AND DEVICES TO AVOID DAMAGE FROM ROOFING SCREWS.
2. PROVIDE SLOTTED CHANNEL SUPPORT TO SPAN ACROSS TRUSSES.
3. COMPLY WITH NEC 300.4(E). PROVIDE AT LEAST 2-INCH SPACING INSTEAD OF NEC MINIMUM 1.5 INCHES.
4. PROVIDE BEAM CLAMPS AT TOP FLANGE OF BEAM OR TOP CHORD OF ROOF JOISTS OR TRUSSES.

A1 CONDUIT AND BOXES BELOW ROOF DECK INSTALLATION DETAIL

SCALE: NOT TO SCALE

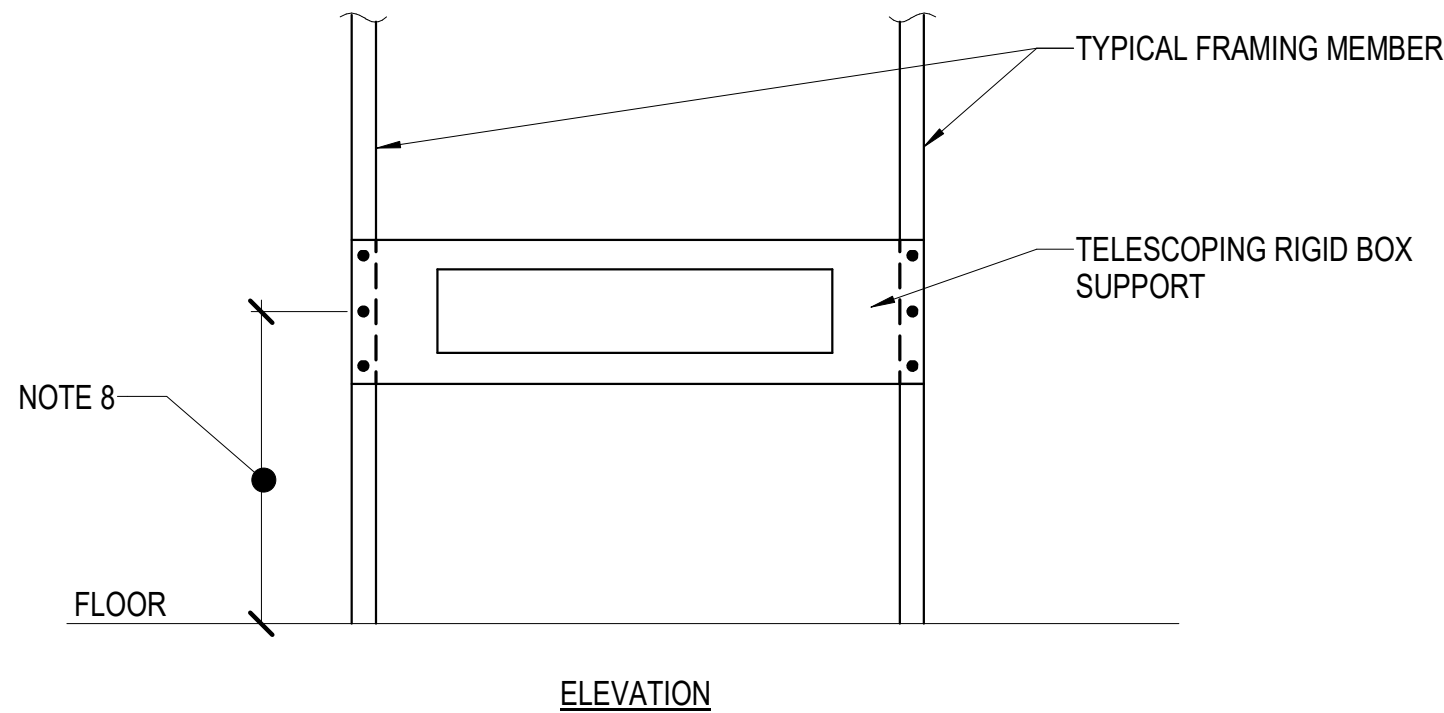


SLOTTED CHANNEL RACK DETAIL NOTES:

1. ANCHOR TO FLOOR AND WALL FRAMING.
2. PROVIDE BOLTED CONNECTIONS BETWEEN STRUT PIECES AND BETWEEN STRUT AND EQUIPMENT.
3. PROVIDE 1/2" CLEAR SLIP JOINT AT VERTICAL SLOTTED CHANNEL TO ALLOW FOR STRUCTURAL DEFLECTION. SECURELY ANCHOR SLIP JOINT TO STRUCTURE. DO NOT SECURE TOP OF VERTICAL SLOTTED CHANNEL TO CEILING OR WALL.
4. PROVIDE ADDITIONAL VERTICAL SLOTTED CHANNEL EVERY 20 INCHES FOR WIDER PANELBOARDS OR CONTROL PANELS OR GROUPS OF PANELBOARDS OR CONTROL PANELS. INCLUDE ADDITIONAL HORIZONTAL SLOTTED CHANNEL PER DETAIL.
5. PROVIDE STEEL SLOTTED CHANNEL SUPPORT FOR MOUNTING PANELBOARDS, CONTROL PANELS, VFD'S, STARTERS AND DISCONNECT SWITCHES.
6. PROVIDE SLIP JOINT TIGHTEN BOLT THROUGH SLOTTED CHANNEL HAND TIGHT. PROVIDE LOCKNUT AND DOUBLE NUT. TIGHTEN NUTS AGAINST EACH OTHER.
7. PROVIDE MANUFACTURER'S NON-METALLIC END CAPS AT CHANNEL ENDS WHERE EXPOSED.

C2 SLOTTED CHANNEL RACK DETAIL

SCALE: NOT TO SCALE



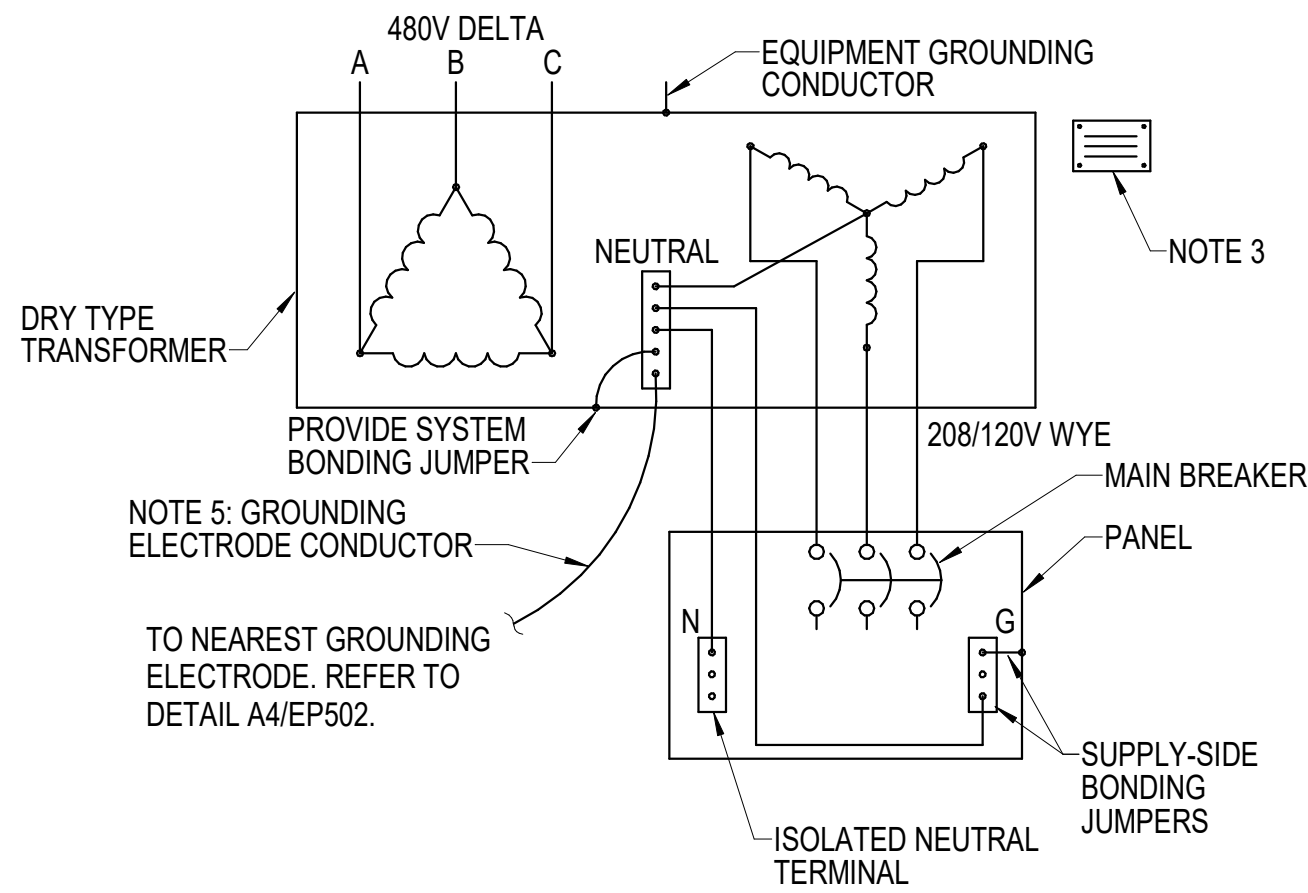
ELECTRIC DEVICE BOX MOUNTING DETAIL NOTES:

1. PROVIDE TELESCOPING RIGID BOX SUPPORT TO LOCATE DEVICES BOXES AT LOCATION INDICATED ON PLANS.
2. PROVIDE AT LOCATIONS WHERE TEMPERATURE SENSORS, POWER ,PHONE, COMMUNICATIONS, SECURITY, AND DATA OUTLETS ARE SHOWN IN THE SAME AREA.
3. PROVIDE AT OFFICE DESKS, SYSTEMS FURNITURE, AND SIMILAR LOCATIONS.
4. PROVIDE TO LOCATE LIGHT SWITCHES, FIRE ALARM PULL STATIONS, RECEPTACLES, EMERGENCY SHUT DOWN SWITCHES AND SIMILAR ITEMS ADJACENT TO DOOR FRAMES.
5. PROVIDE WHERE DEVICES ARE INDICATED AS BEING "CENTERED" WITH RESPECT TO OTHER BUILDING FEATURES.
6. PROVIDE TO LOCATE DEVICES AS INDICATED ON ELEVATIONS. REFER TO AE SHEETS.
7. PROVIDE IN LOCATIONS WHERE FRAMING MEMBER LOCATIONS, SPACING OR QUANTITY IS NOT SUFFICIENT TO LOCATE DEVICES OR GROUPS OF DEVICES AS INDICATED.
8. COORDINATE WITH FRAMING INSTALLER TO ADJUST WALL STUDS FOR CENTERING DEVICE BOXES. PROVIDE ADDITIONAL FRAMING STUD AT LOCATIONS WHERE FRAMING SPACING IS ADJUSTED.
9. DEVICE BOXES TO BE BACK MOUNTED ON PLATE. DEVICE RAISED RINGS TO BE FRONT MOUNTED ON PLATE.
10. VERIFY MOUNTING HEIGHTS AT EACH LOCATION WITH CONTRACTING OFFICER.

A3 ELECTRICAL DEVICE BOX MOUNTING DETAIL

SCALE: NOT TO SCALE

ET502



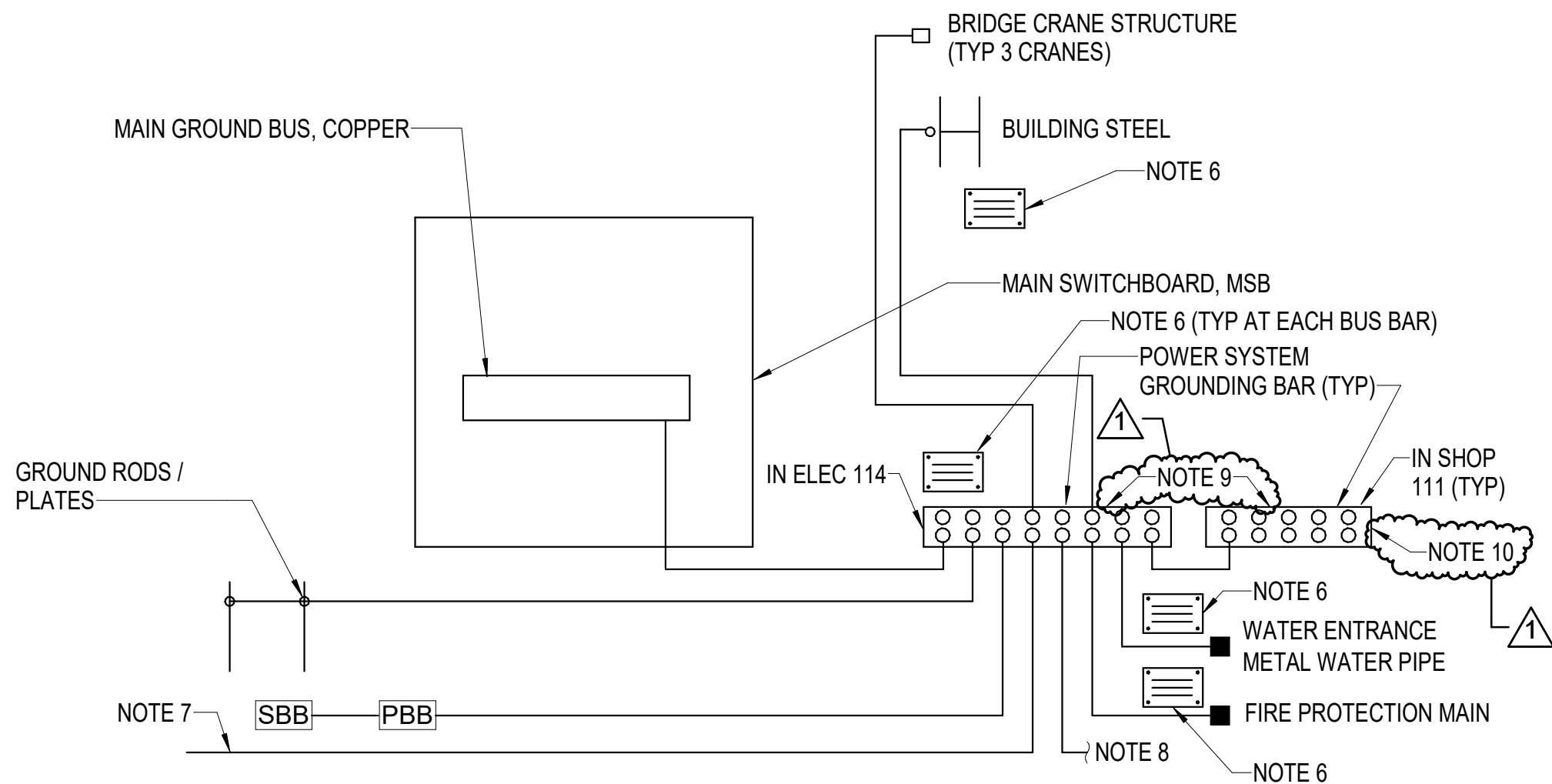
TRANSFORMER GROUNDING DETAIL NOTES:

1. PROVIDE AT DRY-TYPE TRANSFORMER.
2. PROVIDE GROUNDING ELECTRODE SIZED PER NEC FOR TRANSFORMER SIZE AND SECONDARY PANELBOARD MAIN CIRCUIT BREAKER.
3. PROVIDE ENGRAVED INFORMATIONAL LABEL AT TRANSFORMER. INCLUDE SOURCE, TRANSFORMER DESIGNATION, LOAD SERVED, VOLTAGE, AND KVA RATING. PROVIDE ENGRAVED INFORMATIONAL LABEL AT GROUNDING ELECTRODE CONNECTION LOCATION. COORDINATE WITH CONTRACTING OFFICER.
4. PROVIDE SECONDARY CONDUCTORS AS PER NEC FOR SEPARATELY DERIVED SYSTEM.
5. PROVIDE EMT WITH ANTI-CHOKE BONDING CONNECTORS AT EACH END OF GROUNDING ELECTRODE CONDUIT IN ACCORDANCE WITH NEC 250.64(E)(1).

C4 TRANSFORMER GROUNDING DETAIL

SCALE: NOT TO SCALE

EP501



ELECTRIC SERVICE GROUNDING DETAIL NOTES

1. COMPLY WITH NEC, SPECIFICALLY NEC 250.50 AND 250.52.
2. PROVIDE THERMAL WELDED OR IRREVERSIBLE COMPRESSION CONNECTIONS.
3. MINIMUM CONDUCTOR SIZE TO GROUNDING ELECTRODES MUST BE #4/0 COPPER.
4. GROUND PLATE CONNECTION MUST BE UL LISTED, SUITABLE FOR DIRECT BURIAL, THERMAL WELD.
5. WATER PIPE CLAMPS MUST BE BRONZE, UL LISTED FOR WATER OR GAS PIPE SIZE AND INCLUDE LUG WHICH IS UL LISTED FOR GROUNDING ELECTRODE CONDUCTOR SIZE.
6. PROVIDE ENGRAVED WARNING SIGN. INTENDED TEXT: "WARNING: GROUNDING SYSTEM CONNECTION - DO NOT REMOVE".
7. PROVIDE GROUND WIRE WITHIN CONCRETE ENCASED DUCTBANK.
8. FIELD VERIFY AND PROVIDE CONNECTION TO EXISTING BUILDING GROUNDING SYSTEM.
9. SIZE IN ACCORDANCE WITH THE IMMEDIATE APPLICATION REQUIREMENTS AND WITH CONSIDERATION FOR FUTURE GROWTH.
10. PROVIDE AT EACH LOCATION INDICATED IN SHOP 111. PROVIDE A SEPARATE GROUND WIRE FROM EACH GROUND BUS BACK TO THE SERVICE GROUND ("STAR PATTERN").

A4 ELECTRIC SERVICE GROUNDING DETAIL

SCALE: NOT TO SCALE

EP102, EP502

03/16/2026 10/07/2025 DATE		RSB RSB APR
AMENDMENT 009 FINAL SUBMISSION MARK DESCRIPTION		
APPROVED		
FOR COMMANDER NAVFAC		
ACTIVITY		
SATISFACTORY TO DATE		
DESIGNER	HRM	
DRAWN BY	BPD	
CHECKED BY	DCL	
DESIGN MANAGER	T. CARLETON	
PROJECT MANAGER	L. LYONS	
HEADLINE		
FIRE PROTECTION	S. RUCINSKI	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME		
B60 REPAIRS		
ELECTRICAL POWER DETAILS		
PROJECT NO. 1740579		
NAVFAC DRAWING NO.		
SHEET	256	OF 282
EP502		
DRAWING REVISION: 14 SEP 2023		